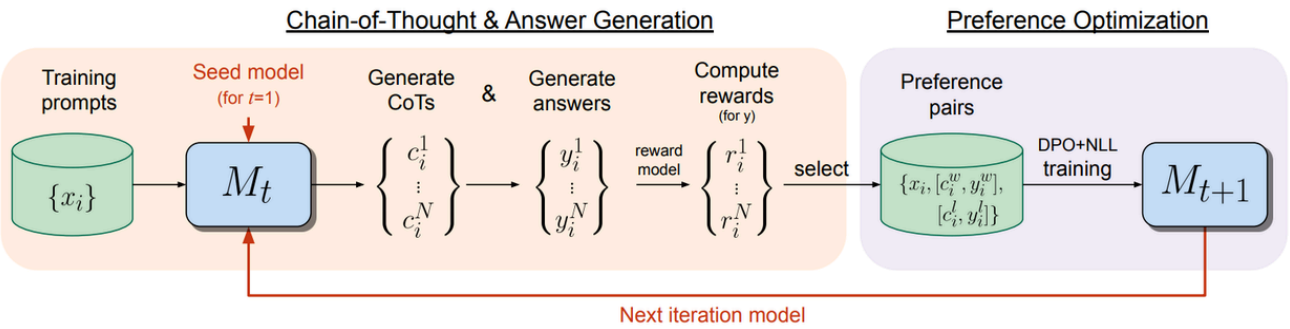
Iterative Reasoning Preference Optimization

> #DPO #RL #Finetune
>
> *It is recommended to read this one together with [offline-online RL]*



- What problems does this framework solve?
  - Online RL algorithms, such as PPO and GRPO, are believed to have a better effect compared to the off-policy algorithms such as DPO. However, they are significantly more expensive.
  - One way to balance the expense and the effect is via an iteratively updating DPO.
- What does this framework do?
  - Sample a set of prompts from the dataset.
  - Let the current policy model M_c generate a set of CoT and answers $\{(c^i, y^i)\}$.
  - Let the reward model validate the answer ONLY, not including the CoT.
  - Select one answer from the correct answer set as the positive response y_w. Select one answer from the wrong answer set as the negative response y_l. Construct the DPO +- pair set.
  - Run optimization with:

$$L = \underbrace{-\log\sigma\left(\beta\log\frac{\pi_\theta(y_w|x)}{\pi_{ref}(y_w|x)} - \beta\log\frac{\pi_\theta(y_l|x)}{\pi_{ref}(y_l|x)}\right)}_{\text{DPO}} + \alpha \underbrace{\left(-\frac{\log\pi_\theta(y_w|x)}{|y_w|+|y_l|}\right)}_{\text{"absolute" optimization (NLL)}}$$

    - Here NLL regulation is similar to the SFT objective function, as it is observed that DPO may cheat by **slightly decreasing the possibility of** y_w(or not change) and **greatly decreasing the possibility of** y_l **to minimize the DPO loss.** To ensure the robustness and stability, we add the NLL term to let the model maximum the "absolute" possibility of the correct answer y_w.
  - After that, we have optimized model M_{c+1}. With it, replay the process.
- Experiment Results
  - Significant Effect: Using only the original training set (without introducing external data), Llama-2-70B achieved accuracy improvement from 55.6% to 81.6% on the GSM8K math task (after 4 rounds of iteration). Substantial gains were also observed on the ARC-Challenge and MATH datasets.
  - Necessity of NLL: Comparative experiments demonstrate that without NLL loss (using only standard DPO), model performance falls below that of simple supervised fine-tuning (SFT), confirming the critical role of the DPO+NLL strategy in inference tasks.
  - Iteration Effectiveness: Performance improves with increasing iterations, though gains gradually

plateau.

- My Thought
  - Compared to *Bridging Offline and Online*, this framework updates the reference model after the dataset is finished instead of after a pre-set number of steps. This may lose some agility of the step-based synchronization.
  - However, *Bridging*... did not include the NLL term that stabilizes the finetuning.
  - It is a way to optimize the framework by combining these two together: let the user choose the frequency of synchronization while using the updated training objective with NLL.
  - Also, the reward can be compound: we both validate the CoT and the result to avoid "correcting by accident". Plus, a non-binary reward can be used to distinguish the answer further, and we select the best answer in the correct set and the worst answer in the wrong set to construct the +-pairs.
- Work Cited
  - Paper
  - Chat History
  - NotebookLM